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Realism and the wavefunction

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Abstract

Realism—the idea that the concepts in physical theories refer to 'things' existing in the real world—is introduced as a tool to analyse the status of the wavefunction. Although the physical entities are recognized by the existence of invariant quantities, examples from classical and quantum physics suggest that not all the theoretical terms refer to the entities: some terms refer to properties of the entities, and some terms have only an epistemic function. In particular, it is argued that the wavefunction may be written in terms of classical non-referring and epistemic terms. The implications for realist interpretations of quantum mechanics and on the teaching of quantum physics are examined.

1. Introduction

For Weinberg, '*wavefunctions are real (...) because it is useful to include them in our theories*' [1][1]. Popper on the other hand, although reluctant to argue about words, believed something is real provided it is '*'kickable' and able to kick back if kicked*' [2]. There is a persistent confusion in the debate on the reality of quantum systems, on what a quantum object is, and even whether there are such things as quantum objects at all. This is not new—the debate between the 'Founding Fathers' of 20th century physics is plagued with misunderstandings arising because of alternative or different meanings conveyed by key words such as realism, determinism, probability, causality, etc. It may be noted that still today, it is widely believed that the defeat of Einstein's position has rang the knell of realism (whereas Einstein's realism is of a very specific kind), or conversely that the de Broglie–Bohm version of quantum mechanics is the only one compatible with realism (whereas history of science taken by and large indicates that the more a theory contains *ad hoc* elements unobservable in principle, the less it refers to the real world).

Concurrently, recent investigations on the students' understanding of quantum mechanics have all highlighted profound conceptual problems. But depending on the authors' conceptions on the status of quantum physics vis-à-vis reality, opposite conclusions have been drawn, eg from the need to '*reconcile non-visualizable quantum physics with visualizable classical physics*' [3] to prescription of '*avoiding reference to classical physics*' [4].

Part of the above-mentioned problems are not so much related to what realism *is* (there is of course no general agreement among philosophers on this point), but rather to establishing

[1] This is in fact Scrooge's opinion, that Weinberg strangely portraits as a realist.

clear-cut categories helpful in the analysis of quantum mechanical reality. Note that the very nature of the physical reality described by quantum mechanics is still today a matter of personal taste or of philosophical prejudice. The *ontological* basis (i.e. what the world is made of) underlying quantum mechanics depends on the meaning given to the theoretical terms: are these terms real, or are they just *epistemic* (i.e. purely knowledge related) tools? Our aim here is to draw some simple conceptual distinctions concerning realism. More specifically, we will analyse the relation between quantum mechanics and physical reality from the standpoint of reference. Present-day scientific realism is generally based on one or other version of a theory of reference, by which theoretical terms are assumed to refer to entities existing in the real world (section 2). This does not entail, however, that *every* theoretical term refers to an existing physical entity. We will argue in sections 3 and 4 that the relationship between the theoretical terms and the physical entities is a relevant starting point to investigate the status of the theoretical terms. We shall take examples of referring and non-referring theoretical terms from classical mechanics, and examine the situation in quantum mechanics; in particular, we will consider whether the wavefunction can be said to be an objective physical system, as has been recently suggested [5]. The implications of our analysis for realist interpretations of quantum mechanics and on the students' understanding of quantum phenomena will be detailed in section 5.

2. Realism

As mentioned in the introduction, there is no agreement among the philosophers of science as to what realism is. In fact, there are very different sorts of realism, and it is not our aim here to review these (see e.g. [6], and [7] and references therein). Here, we will understand by realism not so much the hypothesis that there is an autonomous external world known as 'physical reality' (the overwhelming majority of physicists would agree with this assumption[2]) but rather the stronger idea that the concepts of physical theories refer to 'things' existing in the real world, i.e. physical theories and concepts are more than convenient manners of organizing the data obtained from observation. Indeed, if that were to be the case, that is if physical theories and concepts spoke of objects that do not really exist, then, as Putnam put it, only a miracle could account for the success of science [10]. Note that although the putative referents exist independently of the subject, their relationship to a given theoretical term is not ascertained. Firstly, a theoretical term does not necessarily refer to a real object (which will be called hereafter a 'physical entity'), but may refer to a *property* of an object, to collective properties of an assembly of objects through model-reference, or may not refer at all (examples will be given below). Most importantly, even referring terms are approximate and revisable representations of physical reality. Thus, although in successive theories theoretical terms do change, they may still refer to the same physical entity, provided the description given by the earlier theory is reasonably modified so that the entity conserves its basic roles and properties: a new, more accurate theory brings in novel features, but overlaps with the previous theory on a large domain over which the older theory was known to be accurate. The typical example is Newtonian mechanics, which is still employed in many applications though it is known to be valid only in the non-relativistic limit. This does not mean however that the older theory is necessarily a limiting case of the novel one; for example Franklin concludes, in his historical study [11], that neutrinos retain to this day much of the original properties they had when Pauli originally suggested the existence of such particles, although their currently known properties (they come in three different varieties, and have helicities) do not make them collapse to Pauli's neutrinos.

A few comments are in order. Our assumptions do not entail a correspondence theory of truth, by which the theoretical terms would be endowed with a direct mapping unto physical reality. As a matter of fact, the ontological perspective arising from a theory depends on

[2] There are some well known exceptions however, such as Wigner [8] or Wheeler [9].

whether (and which) theoretical terms are given ontological reference. This last point has always been discussed within the realist perspective and is intimately related to what should be a realist account of truth[3]. We also see that a realistic perspective does not necessarily call for a 'pictorial' representation given in terms of intuitive or familiar categories known to be useful in everyday life; in other words, there is no warrant that an underlying ontology to a successful theory can be spelled out in these categories. And it would not be surprising for such pictorial representations to fail for ontologies concerned with scales far away from the biological realm where our common concepts have originated.

We finally wish to make some observations on Einstein's figure as personifying realism. Einstein's mature position turns out to be at odds with the previous remarks. Probably a remnant of his early Machian heritage, Einstein starts constructing physics from the primary sensory perceptions we experience, but insists into going beyond this phenomenological perspective: physics is defined as a '*logical system of thought*' endowed with a uniform logical basis [13]. He is not so much concerned about whether the theoretical terms refer, but rather about whether a theory is empirically adequate (which according to him is the main justification of the theory). As regarding what type of elements an acceptable theory must contain, Einstein does require the validity of a certain number of concepts from everyday experience (causality, space-time representation), because for him, the miracle is that the world is comprehensible, and his main concern is for theories to give a plausible and intuitively understandable account compatible with the available experimental data. This position was labelled by Fine 'motivational realism' [14], because its role is to give a sense to Einstein's commitment to science, whereas stricto sensu, the epistemological positions are closer to constructive empiricism than to realism.

3. Physical entities and invariance

There undoubtedly are, for most physicists, real entities. Those are detected and manipulated every day in research or development laboratories. The properties of these physical entities are given within a physical theory—but do not entirely depend on the theory[4]. Traditionally, if not intuitively, an entity has a certain number of depictable properties and its evolution can be followed in space and time. Some of these properties are contextual, other are invariants. In classical (including relativistic) mechanics the contextual (usually frame-dependent) properties are related by frame transformations; the invariants have the same value, independently of the transformations. A property may then unambiguously be ascribed to an entity, relative to a given reference frame, or equivalently, the invariants under the given group of transformation may be constructed. This is why Max Born insisted that '*the idea of invariant is the clue to a rational concept of reality*' [16].

This process is blurred in quantum mechanics. Quantum mechanically, an object cannot generally be endowed with the properties it is supposed to carry (this happens in compound systems, or for identical particles)—a situation known as the problem of objectification. And only the time-evolution of the wavefunction of the physical system in configuration space can be followed. Moreover, the wavefunction determines at most mean values in conjunction with an observable, and probability amplitudes when further associated with a definite quantized value (eigenvalue) of the observable. Under general assumptions (without, however, introducing additional hidden variables) it is not possible to attribute an individual *eigenstate* of an observable to the system prior to a measurement (strong objectification) nor even an

[3] These discussions have over the years grown to incorporate sophisticated technical arguments (see [12] for an introduction).

[4] The charge, mass or magnetic moment of the electron does not change when considered within the Bohr–Sommerfeld theory or within the wave mechanics arising from the Schrödinger equation. The Dirac equation further adds a new property (spin) to the electron, which previously needed to be incorporated in an *ad hoc* manner in the non-relativistic treatment. Other properties which were subsequently discovered, such as the electron's role in electroweak interactions, are taken into account by new theories which encompass both the old and the novel properties.

eigenvalue of an observable to the system (weak objectification) [15] because observables do not commute and wavefunctions interfere (or put differently, because of the wave character of the particles): there is no relation between the result of a measurement and the one that would have been obtained from measuring another observable in a different experimental setup, these measurements being mutually exclusive and the result of a measurement not being known with certainty. Hence this additional dependence on the experimental setup (that is, on the projection basis in abstract spaces) which brings in a novel type of hyper-contextuality unknown in classical mechanics.

However, the wavefunction is invariant relative to the decomposition in abstract Hilbert space, i.e. a unitary frame transformation between two projection basis corresponding to two different observables does exist. This 'invariance' has been taken as an argument for the objectivity of the wavefunction itself—the outcome of a measurement then appears as a contextual property of an 'objective' physical system [5]. Notwithstanding, it must be recognized that invariance does not entail reference. Furthermore, realism calls for more than a symbolic transposition or representation: hence the problem of knowing what the wavefunction refers to (see section 4). Note that quite independently from the wavefunction's precise role, invariant quantities are used in quantum mechanics to define the natural kinds, that is to identify the physical entities. An electron is characterized by its mass, charge, spin, whatever the contextual measurement (as a matter of fact, these properties are indirectly measured from different experimental setups). The elementary particles are distinguished by a set of invariant properties. This is also true of compound systems as a whole such as an atom, although further variables associated with the degree of excitation appear. It is thus clear that these invariants are properties of the physical entities that are manipulated experimentally.

4. Theoretical terms and reference

4.1. Classical mechanics

The problem of the reference of theoretical terms seems at first sight straightforward in classical mechanics: either the objects exist, or they do not. A simple well known example is given by the debate on the existence of the putative atoms postulated by the kinetic theory of gases towards the end of the 19th century: were these atoms mere theoretical tools which saved the phenomena, as the instrumentalists argued, or were they real but at-the-time unobserved particles? A less simpler case concerns the existence of the ether, which was postulated to exist because it was needed by the ontological picture, although there was not any compelling theoretical term which referred to it. A much more involved example can be found in the criticism of the Newtonian concept of force, which was much criticized from its inception onwards as being obscure, while reference to less obscure 'active principles' was looked for. These examples illustrate the subtle interplay between a physical theory, here classical mechanics, and the ontological basis that goes with it, in the construction of an interpretation.

In classical mechanics, the space–time (Newtonian) formulation gives rise to a simple reference procedure: the fundamental theory gives the properties of material points i as evolution of the position $\boldsymbol{x}_i(t)$ and velocity $\boldsymbol{v}_i(t)$ of those points interacting through a function $V(\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{v}_1, \boldsymbol{v}_2, \ldots)$. The time evolution is given by a differential equation of the type

$$m_i \frac{\mathrm{d}^2 \boldsymbol{x}_i}{\mathrm{d}t^2} = -\boldsymbol{\nabla}_i V + \frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial V}{\partial \boldsymbol{v}_i} \tag{1}$$

which is readily amenable to an interpretation in terms of the action of a force on the dynamics of the material points. Larger or complex systems are investigated within models based on the fundamental theory, involving the introduction of approximations; we speak of *model-reference*. The relevant theoretical terms then refer to properties of physical entities (usually some collective properties) through the model, which by definition is itself an approximation. In statistical theories however, the terms do not refer to properties of the physical entities: a statistical distribution gives information related to an ensemble of systems. It is an *epistemic* tool which at best refers to collective properties of a finite number of identical systems, not to a property of an individual system.

Now, the most powerful and elegant formulation of classical mechanics based on the theory of canonical transformations employs theoretical terms which have a much more abstract and complicated relation to the physical entities. An original set of independent coordinates and momenta (q_i, p_i) is transformed to a new set (Q_i, P_i) by means of a generating function [17]. When all the Q and P are chosen to be constants of motion, the generating function is the classical action $S(q_i, P_i, t)$ obtained by solving the Hamilton–Jacobi equation

$$H\left(q_i, \frac{\partial S}{\partial q_i}, t\right) + \frac{\partial S}{\partial t} = 0, \tag{2}$$

where H is the Hamiltonian expressed in terms of the independent coordinates, and the original momenta are retrieved by using

$$p_i = \frac{\partial S}{\partial q_i}. \tag{3}$$

Of course, other generating functions expressed in terms of different sets of independent coordinates may be chosen (e.g. the function labelled $F_4(p_i, P_i, t)$ by Goldstein [17]; in this representation, the p_i are the independent coordinates, and the positions are retrieved by the relation $q_i = -\partial_{p_i} F_4$). Theoretical emphasis is put on canonically invariant quantities, such as the Poisson bracket of two quantities $a(q, p)$ and $b(q, p)$, defined by

$$\{a, b\} \equiv \sum_i \frac{\partial a}{\partial q_i}\frac{\partial b}{\partial p_i} - \frac{\partial a}{\partial p_i}\frac{\partial b}{\partial q_i}. \tag{4}$$

The effect of a transformation can then be written in terms of Poisson brackets; for example the time evolution of $a(p, q)$ is given by

$$\frac{\mathrm{d}a}{\mathrm{d}t} = \{a, H\} + \frac{\partial a}{\partial t}, \tag{5}$$

where H is now expressed in terms of p and q.

What does the Poisson bracket or the classical action refer to? There is clearly no place for them in the ontology underlying classical mechanics; these terms are not properties of physical entities. Relative to the ontology of classical mechanics, they are epistemic terms from which we can extract the dynamics of the system (the action—a non-local quantity—gives all the possible trajectories compatible with the mechanical system, the bracket gives the evolution generated by an infinitesimal canonical transformation). The propagation of the action in configuration space can be studied for its own sake (for example, the surface of constant S forms a wavefront normal to the trajectories propagating with velocity E/p). But this does not make the action a referring term, although we know how to associate this term with the trajectories of physical entities.

4.2. Quantum mechanics

The main problem in understanding quantum mechanics is centred on the meaning of the wavefunction. Ontological interpretations hinge on the task of referring the wavefunction to some recognizable physical phenomenon, and there is clearly no agreement on this point. Schrödinger's original position, abandoned soon afterwards, was to envision the wavefunction as existing in real space. The well known 'many worlds interpretation' is a consistent manner of taking the wavefunction at face value. A more elaborate form of combining projective decompositions of the wavefunction evolution is achieved by the consistent histories interpretation; however, the ontological status of the histories (each one of them is consistent, but two histories are generally incompatible accounts of the same phenomena) is open to question. In the de Broglie–Bohm account of quantum mechanics, the wavefunction is referred to a real physical field existing in configuration space; this space therefore acquires an ontological existence. More recently, the introduction of non-destructive measurements has lead certain authors [18] to ascribe to the wavefunction the role of referring to a time-averaged property of a particle.

To tackle the problem of the reference of the wavefunction, let us start by recalling that there are three well-established connections between quantum and classical theoretical terms[5]. First, there is the formal correspondence between a classical Poisson bracket and the quantum mechanical commutator, a procedure known as canonical quantization. For example the classical relation $\{q_i, p_j\} = \delta_{ij}$ has the quantum counterpart $(\mathrm{i}\hbar)^{-1}[\hat{q}_i, \hat{p}_j] = \delta_{ij}$, where the hat denotes an operator, and equation (5) has the counterpart

$$\frac{\mathrm{d}\hat{a}}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}[\hat{a}, \hat{H}] + \frac{\partial \hat{a}}{\partial t}, \tag{6}$$

that is the evolution of a quantum mechanical operator $\hat{a}(t)$ in the Heisenberg picture. Second, the path-integral approach gives the transition amplitude of the evolution operator in the time interval $t_f - t_0$ as

$$\langle q_f, t_f | q_0, t_0 \rangle = \int \mathcal{D}(q) \exp\left[\mathrm{i} \int_{t_0}^{t_f} \mathrm{d}t \, L(q, \dot{q})/\hbar \right] \tag{7}$$

where the integral of the Lagrangian L is of course $S(q_f, t_f; q_0, t_0)$, the classical action. Finally, the so-called WKB theory gives the semiclassical approximation of the wavefunction as

$$\psi(q_f, t_f) = A(q_0, t_0) \left| \det \frac{\partial q_0}{\partial q_f} \right|^{1/2} \exp[\mathrm{i}S(q_f, t_f; q_0, t_0)/\hbar - \mu\pi/2]; \tag{8}$$

A^2 represents the classical probability density and μ is the Maslov index keeping track of the caustics (the points in phase-space where the semiclassical approximation is not valid) along the trajectory.

Note that in order to obtain equations (7) and (8) assumptions which are seldom mentioned and which are not compelling within quantum mechanics must be made. In equation (7), although the different variables q_k that appear in the measure $\mathcal{D}(q)$ are *independent* variables of integration, the classical action is obtained only by forcefully identifying $q_{k+1} - q_k$ with $\dot{q}_k \, \mathrm{d}t$ [19]. In an analogous manner S is only one of the infinite number of phases yielding the semiclassical wavefunction (8) (quantum mechanics only imposes that all these phases obey a nonlinear equation [20], provided the amplitude A changes accordingly). These stronger assumptions must be made if any connection between quantum mechanical and classical are to be made at all. These assumptions are known to be consistent; for example, in many systems for which $\hbar$ can be scaled, preferential propagation of the wavefunction along classical periodic trajectories has been observed as $\hbar \to 0$, a finding which has sparked progress in many branches of physics [21].

What is striking is that the quantum mechanical relations are made up with quantities that classically have an *epistemic non-referring* interpretation[6]. The operators alone are not sufficient to obtain the dynamics; they must be associated with a wavefunction to obtain eigenvalues or mean values. The wavefunction is generally obtained from the eigenfunctions of the Hamiltonian, and in any case depends on the interactions between the particles: for example, the wavefunction of, say an electron in a magnetic field, is quite different from the wavefunction of an electron in a Coulomb field, although in both cases the wavefunction is

[5] We leave aside the phase-space formulation of quantum mechanics of the Weyl–Wigner–Moyal type, where the connection between quantum and classical terms is more complex.

[6] There is of course no warrant that the same interpretation needs to hold within quantum mechanics. In fact, there are historical examples and counter-examples: Ptolemy's epicycles and crystalline spheres never came to refer to anything, whereas the energy, which was regarded as an abstract mathematical tool, became a central referring property by the end of the 19th century.

related to the same physical entity that has the same invariant properties. The fact that the wavefunction conveys different dynamical information while supposedly referring to the same object is again in favour of an epistemic interpretation. The wavefunction invariance (in Hilbert space) argument mentioned in section 3, according to which the objectivity of the wavefunction is increased by this invariance, was seen to have its counterpart in classical mechanics, where different representations for the abstract and non-referring generating functions can be chosen. Other arguments which render difficult the task of referring the wavefunction include the unchallenged probability postulate and thus the need for a normalization factor, the irrelevance of a global phase in the wavefunction and the non-invariance under Galilean transformations of the local wavelength of the wavefunction. The most problematic point, however, remains the reduction of the wavefunction during a measurement process. It seems unlikely that any decisive progress on the meaning of the wavefunction will be made before a satisfying solution to the measurement problem will be found.

5. Implications

We have argued that within a realist perspective, it is not compelling to refer the wavefunction to a physical entity, since not all theoretical terms refer to physical entities or their properties. We examine the implications on the interpretation and the teaching of quantum mechanics.

5.1. Interpretation

Whereas empiricists and instrumentalists consider quantum mechanics as a masterpiece to validate their arguments, realist interpretations have been obscured by two points. First, a certain number of preconceived epistemological constraints that were imposed as basic requirements in any realistic account (an example is Einstein's position, as sketched in section 2); for example, it is often thought that realism imposes an isomorphism between the theoretical terms and nature (e.g., objectivity, understood in the precise sense of relating a theoretical term with an element of reality with unit probability—as it appeared in the EPR paper [22]—is believed to be a basic prerequisite for any theory compatible with realism). The upshot is that these epistemological constraints aim at bypassing any epistemic barrier between our theories and reality, thus implicitly assuming a pre-structured reality and a correspondence theory of truth.

The second point concerns a preconceived ontology; a realist account is often thought to be necessarily spelled out in apparently familiar terms, although there is a price to pay because the ontology must be modified to account for the novel phenomena. A demonstrative example is the de Broglie–Bohm approach to quantum mechanics [23]: a wave—a real physical field propagating in configuration space—guides the particles along in principle unobservable trajectories. Originally devised to restore a continuity between classical descriptions and quantum phenomena—it was later claimed that it would avoid '*arbitrary dichotomies . . . between evident (macroscopic) realism and quantum (microscopic) nonrealism*' [24]—, Bohmian mechanics has been led to ascribe to the putative entities a haul of counter-intuitive properties: particles are detected even though no trajectories go nearby the detector, properties such as the mass of the particles are delocalized, configuration space has sometimes been claimed to be more fundamental than our 3 + 1D space–time, classical trajectories are seldom obtained in the classical limit. Such peculiarities are given *ad hoc* assumptions if necessary so as to yield the standard or experimental results. In view of this situation, it might appear as a paradox for Bohmian mechanics to be coined *the* interpretation of quantum mechanics compatible with realism, even sometimes as the main counter-argument used by many realist philosophers with an interest in the interpretation of quantum mechanics [25, 26][7].

[7] This forms part of a more general tendency some philosophers have which is to use any interpretation giving the wavefunction an ontological existence as an argument for realism.

The consistent histories approach [27] offers a quite different perspective regarding the interplay between epistemological and ontological constraints. From our point of view, the main feature comes from the existence of alternative and mutually exclusive consistent histories. For example, in a beamsplitter (this example is discussed in [28]), the neutron, at some intermediate time t_1 before being detected, is in either of the two channels if it is described by two different histories of a certain family; a history belonging to another family states that at t_1 the wavefunction is in a superposition state. What can be the ontological value of this 'many-picture' formalism? Each history taken individually makes sense from an epistemic standpoint (and the history thereby allows us to intuit the physical evolution of the system in-between observations), but *any* of the alternative and complementary pictures may be taken as valid. The properties of the physical phenomena are thus context dependent through the history that is chosen for the description, and in this respect, the consistent histories interpretation does not shed new light on the properties and objectification problems briefly mentioned in section 3. Maybe our brains lack the categories in which reference could be expressed (we would then be in the presence of an internal biological epistemic barrier) and the problem of objectification can therefore be dismissed: there are real physical entities, but as far as we can know, their properties can only be described relative to a given history (compatible with a given experimental setup). However, we noted there are invariant, context-independent properties, and we suggested that the wavefunction was 'made-up' of classically non-referring terms. Moreover, an actual measurement performed at an intermediate time yields a unique result, and we know that a unique classical macroscopic description arises from the diverse quantum histories. This is why, in our view, the task of constructing an ontological picture from the physical theory hinges on the resolution of the measurement problem—the quantum mechanical description of classical phenomena[8].

We have seen that realism does not entail a correspondence theory of truth by which every element of reality could be mapped onto theoretical terms. Moreover, realism is not concerned by adapting a physical theory to a preconceived ontological basis: as celebrated examples in the history and philosophy of science have taught us, this can only be done by developing a plethora of *ad hoc* hypotheses, increasingly non-referring and unphysical. If, as we have argued, realism is based on referring theoretical terms to physical entities, there is no place for in principle unobservable properties, because observation is the warrant of referentiability, i.e. it is only by repeating and combining observations that the referents can be accessed [30]. Although we know how to recognize real physical entities, through their invariant properties, it is not clear what the furniture of the quantum world is.

5.2. Teaching

If, following Feynman, *nobody* does understand quantum theory (for a more nuanced review of the present situation, see [31]) particular difficulties in the teaching of quantum physics are to be expected. Indeed, recent investigations in the students' understanding of quantum phenomena have not surprisingly diagnosed profound conceptual difficulties [3, 4, 32]. More surprisingly however, opposite conclusions have been drawn from these studies: in one case [4] it is recommended to avoid any reference to classical physics and to dual quantum/classical descriptions, such as the Bohr atom, in favour of statistical interpretations of observed phenomena. In another case [3], it was found that the difficulties arise from the ontological and epistemological status that students ascribe to the theoretical terms, and it was suggested to develop 'mental models' that would reconcile quantum and classical physics. More generally the findings indicate conceptual difficulties that later persist when more advanced material is studied; only specific tutorials based on constructing concepts and '*relating the formalism of physics to the real-world phenomena*' [32] were found to be efficient.

[8] There have been claims (e.g. [27]), that at least for practical purposes, the existence of classical properties is well understood, but there is no agreement even within the consistent histories perspective [29].

If conceptual problems deserve conceptual treatments, it then appears that the specific problems raised by quantum mechanics can hardly be understood without going into a more general inquiry on the relationship between theories and reality. We have argued that an approach from the realist standpoint of reference is well suited in order to understand this relationship. Globally, this suggests that introductory courses to the philosophy of science would help the students in their confrontations with the inevitable conceptual problems that arise when trying to understand what is quantum mechanics and what is quantum reality [33].

6. Conclusion

Realism leads us to believe that physical theories refer to real physical entities. This does not mean that every theoretical term must refer. We have given examples of theoretical terms that have an epistemic non-referring signification in classical mechanics. We have further seen that quantum mechanical theoretical terms can be constructed with these classical epistemic terms. The tension between the invariant properties of the physical entities and the contextual nature of the wavefunction suggests that a realist interpretation will come through with the solution to the measurement problem, rather than by imposing preconceived epistemological or ontological constraints.

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The report by Professor D Lecourt which is currently leading to the widespread introduction of philosophy of science courses in the French scientific universities is available on the Internet (http://www.education.gouv.fr/rapport/lecourt/, in French only)